

Feb 5 2026

Center manifold Theorem

$$\dot{x} = f(x)$$

$$A = \left. \frac{\partial f}{\partial x} \right|_{x=0}$$

↑

Jacobian has "k" eigenvalues with zero real parts

n-k eigenvalues with negative real parts

$$\begin{bmatrix} y \\ z \end{bmatrix} = Tx \quad \text{change of variables:}$$

↙ block diagonal

$$\dot{y} = A_1 y + g_1(y, z)$$

$$[A_1]_{k \times k}$$

$$\dot{z} = A_2 z + g_2(y, z)$$

$$[A_2]_{(n-k) \times (n-k)}$$

↑

block diagonal

$$\begin{bmatrix} A_1 & 0 \\ 0 & A_2 \end{bmatrix} = T A T^{-1}$$

Jordan normal form

- Find $z = h(y)$

In previous example, $h: \mathbb{R} \rightarrow \mathbb{R}$

in general: $h: \mathbb{R}^k \rightarrow \mathbb{R}^{n-k}$

$$w = z - h(y)$$

$$\dot{w} = \dot{z} - \frac{\partial h}{\partial y} \dot{y} = 0 \quad \text{for } w(0) = 0$$

$$= A_2 z + g_2(y, z) - \frac{\partial h}{\partial y} (A_1 y + g_1(y, z)) = 0$$

$$A_2 h(y) + g_2(y, h(y)) - \frac{\partial h}{\partial y} (A_1 y + g_1(y, h(y))) = 0$$

- Find "reduced" system

$$\dot{y} = A_1 y + g_1(y, h(y))$$

- Analyze stability of "reduced" system

$$J = \begin{bmatrix} J_1 & & \\ & J_2 & \\ & & \ddots \\ & & & J_r \end{bmatrix} \quad r \leq n$$

J_i : submatrix of one of the following

$$J_i = \begin{bmatrix} \lambda_i & 1 & 0 \\ & \ddots & \\ 0 & & 1 \\ & & & \lambda_i \end{bmatrix} \quad \text{or} \quad J_i = \begin{bmatrix} a_i & -b_i \\ b_i & a_i \end{bmatrix}$$

$$\lambda_i = a_i + bi$$

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

$$J = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

$$T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \lambda = 1, 1$$

$\dim$ of $N(A - \lambda I)$ = geometric multiplicity of λ

$Av = \lambda v$: v is an eigenvector corresponding to e-value λ

$$\Rightarrow (A - \lambda I)v = 0$$

$$v \in N(A - \lambda I)$$

$$A - \lambda I = \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_A - (1) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$N\left(\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}\right) = \mathbb{R}^2$$

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 1 \\ 0 & 1-\lambda \end{bmatrix}$$

$$\det(A - \lambda I) = (1 - \lambda)^2 = 0$$

algebraic multiplicity = 2

$$\lambda = 1, 1$$

$$A - \lambda I = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} - (1) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$N\left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\right) = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$$

$$\dim N\left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\right) = 1 = \text{geometric multiplicity of } \lambda = 1$$

$$(A - \lambda I)^2 v = 0$$

Ex. $\lambda = 0, 0$ geometric multiplicity = 1

$$J = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

λ : algebraic multiplicity 4

geometric multiplicity 2

$$J = \begin{bmatrix} \boxed{\begin{matrix} \lambda & 1 \\ 0 & \lambda \end{matrix}} & J_1 & 0 & 0 \\ 0 & 0 & \boxed{\begin{matrix} \lambda & 1 \\ 0 & \lambda \end{matrix}} & J_2 \end{bmatrix}$$

$$\text{or } J = \begin{bmatrix} \boxed{\begin{matrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{matrix}} & J_1 & 0 \\ 0 & 0 & 0 & \boxed{\lambda} J_2 \end{bmatrix}$$

$$\dot{x} = f(t, x), \quad x(t_0) = x_0 \quad (1)$$

Existence and uniqueness

Given $f(t, x)$, x_0 , t_0 , does (1) have a solution $x(t)$ and if it is unique

$$DT \quad x_{t+1} = f(x_t) \quad x(t_0) = x_0$$

$$\text{fixed point: } x = f(x)$$

Fixed point Theorems

$$\text{If (1)} \quad U \xrightarrow{T} U \quad \text{Contraction Mapping Theorem}$$

(1) U : closed subset of a Banach space

$$\text{and (2)} \quad |T(x) - T(y)| \leq p |x - y|$$

$$p < 1 \quad \forall x, y \in U$$

then T has a unique fixed point in U

and the solutions of $x_{n+1} = T(x_n)$ converge to this fixed point from any $x_0 \in U$

Recall $x_{t+1} = r x_t (1 - x_t)$

$$T = r x (1 - x) \\ U \xrightarrow{T} U$$

$$\underline{r=1} \quad U = [0, 1] \quad T = x(1-x)$$

$$\underline{r=4} \quad [0, 1] \xrightarrow{4x(1-x)} [0, 1] \\ U \quad U$$

$$\begin{aligned} T(x) - T(y) &= r |x(1-x) - y(1-y)| \\ &= r |(x-y) - (x^2 - y^2)| \\ &= r |(x-y)(1 - (x+y))| \end{aligned}$$

$$< 3r |x-y|$$

$$|1 - (x+y)| < 1 + x + y < 3$$

$$|1 - (x+y)| \leq 1$$

$$r < \frac{1}{3} \Rightarrow p < 1 \quad \text{contraction}$$

f

$$\text{Alternatively, } |T(x) - T(y)| < r |x - y| \quad |1 - 2|$$

or
|| - 0 |

$$< r |x - y|$$

f

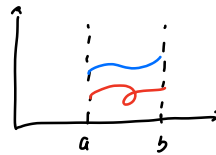
$$r < 1 \Rightarrow p < 1$$

Normed Vector (linear) Space

X : vector space

$|\cdot|$: norm

- $|x| \geq 0 \quad \forall x \in X$
- $|x+y| \leq |x| + |y| \quad \forall x, y \in X$
- $|\alpha x| = |\alpha| \cdot |x| \quad \forall \alpha \in \mathbb{R} \text{ and } x \in X$



Ex.

$C^n[a, b]$: the set of all continuous functions $[a, b] \rightarrow \mathbb{R}^n$ with norm:

$$|x|_C = \max_{t \in [a, b]} |x(t)|$$